PROBLEM STATEMENT #16 | Healthcare & Telemedicine

Remote Patient Vitals Alert & Monitoring App

1. Problem Context & Overview

A continuous monitoring pipeline for post-operative patients that ingests vital telemetry (SpO2, Heart Rate, BP), detects threshold breaches, and escalates emergency alerts through a caregiver matrix.

Target Stakeholders / Actors: *Remote Patient, On-Call Caregiver*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]
Description: The system shall continuously evaluate ingested patient vitals against configured clinical thresholds and flag critical anomalies within 2 seconds.
Sample Acceptance Criteria: Pass: High heart rate (>140 BPM) triggers immediate escalation alert; Fail: Metric spike ignored or delayed > 5s.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]
Description: The telemetry ingestion gateway shall support at least 500 concurrent continuous telemetry streams with 99.99% uptime.
Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**
  >> Exactly 5 Functional Requirements (FR-001 to FR-005)
  >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with
  ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**
  >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**
  >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
  >> One Alternate Flow.